Topological Superconductor

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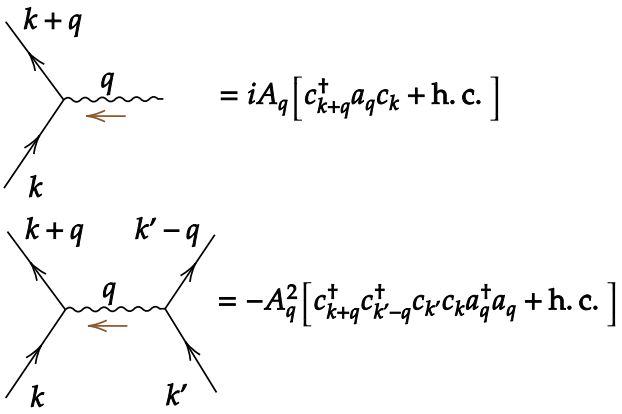


FIG. 1. Microscopic electron-phonon coupling process.

Abstract

SMTI-SC and TSC.

Contents

I. Basics for Superconductor 3

II. SMTI-SC Heterostructure 5

III. Basis Consideration 11

A. Case I: Uniform Zeeman and SC-Pairing 13

B. Case II: Top Zeeman and Bottom SC-Pairing 13

IV. Sau-Lutchyn-Tewari-Das Sarma proposal 14

V. Qi-Hughes-Zhang Model 20

VI. TI Film Model 23

References 23

I. Basics for Superconductor

Superconductor under Bardeen-Cooper-Schrieffer (BCS) theory originates from electron-phonon coupling induced Fermi surface instability in metal. The microscopic Hamiltonian with phenomenological parameters reads

$$H = H_e + H_p + H_{ep} \tag{1a}$$

$$H_e = \sum_k \xi_k c_k^\dagger c_k + \sum_{k,k',q} V_q c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k, \tag{1b}$$

$$H_p = \sum_q \hbar\omega_q \left(a_q^\dagger a_q + \frac{1}{2} \right), \tag{1c}$$

$$H_{ep} = i \sum_{k,q} A_q [c_{k+q}^\dagger a_q c_k + \text{h.c.}]. \tag{1d}$$

The effective Hamiltonian describing electrons comes from integrating phonons out, up to second order as shown in the second diagram in Fig. 1, the effective electron-electron coupling becomes

$$\begin{aligned} \sim \langle H_{ep}^2 \rangle_p &= -\sum_{k,k',q,q'} \langle A_q A_{q'} [c_{k+q}^\dagger a_q c_k + \text{h.c.}][c_{k'+q'}^\dagger a_{q'} c_{k'} + \text{h.c.}] \rangle_p \\ &= -2 \sum_{k,k',q,q'} A_q A_{q'} c_{k+q}^\dagger c_k c_{k'}^\dagger c_{k'+q}^\dagger \langle a_q a_{q'}^\dagger \rangle_p \\ &= -\sum_{k,k',q} \tilde{A}_q^2 c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k, \end{aligned} \tag{2}$$

which is an attractive interaction. Assume that this attraction overwhelms the Coulomb repulsion, which is usually possible in a metal with dense electron gas where Coulomb interaction is effectively screened, we can phenomenologically write down

$$H = \sum_k \xi_k c_k^\dagger c_k - \frac{g}{2} \sum_{k,k',q} c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k \approx \sum_k \xi_k c_k^\dagger c_k - \frac{g}{2} \sum_{k,q} c_{k+q}^\dagger c_{-k-q}^\dagger c_{-k} c_k, \tag{3}$$

where we assume constant coupling and use maximal coupling volume principle. The wavevector q is bounded by phonon dispersion $|\omega_q| < \omega_D$ with ω_D usually Debye frequency.

Generally, the Fermionic operator $c_k = \{c_k^\alpha\}$ contains inner degrees of freedom, correspondingly

$$H = \sum_k \xi_k c_k^{\alpha,\dagger} c_k^\alpha - \frac{g}{2} \sum_{k,q} c_{k+q}^{\alpha,\dagger} c_{-k-q}^{\beta,\dagger} c_{-k}^\beta c_k^\alpha. \tag{4}$$

Introduce (self-consistent) mean-field order parameter, which also represents the superconducting gap, and the electronic correlation respecting Pauli principle

$$\Delta_{\alpha\beta}(k) = g \sum_q \langle c_{k+q}^\alpha c_{-k-q}^\beta \rangle = -\Delta_{\beta\alpha}(-k), \tag{5}$$

and by decomposing $\Delta_{\alpha\beta}(k) = \chi_{\alpha\beta}\Delta(k)$ with χ the spinor matrix for Fermionic inner degrees of freedom (spin, orbital, layer, $\cdots$), we recognize two cases

$$\begin{cases} \chi_{\alpha\beta} = -\chi_{\beta\alpha},\ \Delta(k) = \Delta(-k),\ \text{Singlet} \\ \chi_{\alpha\beta} = \chi_{\beta\alpha},\ \Delta(k) = -\Delta(-k),\ \text{Triplet} \end{cases}, \tag{6}$$

with the former for $s, d, \cdots$ wave pairing while the latter for $p, f, \cdots$ wave pairing. With introducing the pairing, the Hamiltonian becomes a non-interacting one (constant ignored)

$$H_M = \sum_k \xi_k^\alpha c_k^{\alpha,\dagger} c_k^\alpha + \frac{1}{2}\sum_k \Delta_{\alpha\beta}(k) c_k^{\alpha\dagger} c_{-k}^{\beta\dagger} + \text{h.c.} \tag{7}$$

Further introduce

$$\psi_k = \begin{pmatrix} c_k^{\alpha_1} \\ c_k^{\alpha_2} \\ \vdots \end{pmatrix},\ \Psi_k = \begin{pmatrix} \psi_k \\ \psi_{-k}^\dagger \end{pmatrix}, \tag{8}$$

where $\psi^\dagger$ in second definition should be understood as a column vector, we have

$$H_M = \sum_k \psi_k^\dagger \xi_k \psi_k + \frac{1}{2}\psi_k^\dagger \Delta_k \psi_{-k}^\dagger - \frac{1}{2}\psi_k \Delta_k^* \psi_{-k} = \frac{1}{2}\sum_k \Psi_k^\dagger h_{\text{BdG}}(k) \Psi_k, \tag{9}$$

with $(\Delta_k)_{\alpha\beta} = \Delta_{\alpha\beta}(k)$, and

$$h_{\text{BdG}}(k) = \begin{pmatrix} \xi_k & \Delta_k \\ -\Delta_k^* & -\xi_k^T \end{pmatrix}. \tag{10}$$

On the other hand, when α includes spin degrees of freedom, represented by σ, a new basis known as Nambu spinor can be introduced as

$$\Phi_k = \begin{pmatrix} \psi_k \\ (i\sigma_y)\psi_{-k}^\dagger \end{pmatrix}, \tag{11}$$

under which the Hamiltonian equals

$$H_M = \sum_k \Phi_k^\dagger \begin{pmatrix} \xi_k & \Delta_k(-i\sigma_y) \\ -i\sigma_y \Delta_k^* & -\sigma_y \xi_{-k}^T \sigma_y \end{pmatrix} \Phi_k. \tag{12}$$

Notice that two electronic vectors in Nambu basis are time-reversal partners.

II. SMTI-SC Heterostructure

The Hamiltonian of 3-D **bulk** TI[1–5] is represented by the standard modified Dirac's equation[6] on lattice $\mathcal{H}_{TI} = \sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}} H(\boldsymbol{k}) \Psi_{\boldsymbol{k}}$ where

$$H(\boldsymbol{k}) = \sum_\mu \lambda_\mu \sin(k_\mu a_\mu)\alpha_\mu + \left[m_0 - 4t_\mu \sin^2\left(\frac{k_\mu a_\mu}{2}\right)\right]\beta, \tag{13}$$

and given that $\lambda_\mu > 0$ and $t_\mu > 0$, the topology of the system is determined by the low-energy mass term m_0, which drives the system into trivial, weak TI or strong TI with four indices.

An anisotropic 3-D topological insulator **film** with limited number of sites along z direction, by Fourier transformation in $x-y$ plane, reads

$$\mathcal{H}_{\text{Film}} = \sum_{l_z,\boldsymbol{k}} \left(\psi^\dagger_{l_z,\boldsymbol{k}} \mathcal{M}_0(\boldsymbol{k}) \psi_{l_z,\boldsymbol{k}} + \psi^\dagger_{l_z,\boldsymbol{k}} \mathcal{T}_z \psi_{l_z+1,\boldsymbol{k}} + \text{h.c.}\right) + \sum_{l_z,\boldsymbol{k}} \psi^\dagger_{l_z,\boldsymbol{k}} H_\parallel \psi_{l_z,\boldsymbol{k}}, \tag{14}$$

with

$$H_\parallel = \lambda_\parallel[\sin(k_x a)\sigma_x\tau_x + \sin(k_y b)\sigma_y\tau_x], \tag{15}$$

and $\mathcal{M}_0(\boldsymbol{k}) = M_0(\boldsymbol{k})\sigma_0\tau_z = [m_0(\boldsymbol{k}) - 2t_\perp]\sigma_0\tau_z$, $\mathcal{T}_\mu = t_\mu\beta - i\frac{\lambda_\mu}{2}\alpha_\mu$, where

$$m_0(\boldsymbol{k}) = m_0 - 4t_\parallel\left(\sin^2\frac{k_x a}{2} + \sin^2\frac{k_y b}{2}\right). \tag{16}$$

Note that we have set $t_x = t_y = t_\parallel$, $t_z = t_\perp$, $\lambda_x = \lambda_y = \lambda_\parallel$, $\lambda_z = \lambda_\perp, a = b$. The Fermionic vector $\psi = (c_{1\uparrow}, c_{2\uparrow}, c_{1\downarrow}, c_{2\downarrow})^T$, corresponding to $\sigma\tau$ representing spin $\otimes$ orbital.

The **equivalently effective model**[7, 8] is obtained by:
(i) Separate Hamiltonian at $\boldsymbol{k}$ as

$$\mathcal{H}_{\text{Film}}(\boldsymbol{k}) = \mathcal{H}_{1d}(\boldsymbol{k}) + \mathcal{H}_S(\boldsymbol{k}) \tag{17}$$

with

$$\mathcal{H}_{1d}(\boldsymbol{k}) = \sum_{l_z}\left(\psi^\dagger_{l_z,\boldsymbol{k}}\mathcal{M}_0(\boldsymbol{k})\psi_{l_z,\boldsymbol{k}} + \psi^\dagger_{l_z,\boldsymbol{k}}\mathcal{T}_z\psi_{l_z+1,\boldsymbol{k}} + \text{h.c.}\right), \tag{18a}$$

$$\mathcal{H}_S(\boldsymbol{k}) = \sum_{l_z}\psi^\dagger_{l_z,\boldsymbol{k}} H_\parallel \psi_{l_z,\boldsymbol{k}}. \tag{18b}$$

(ii) Solve $\mathcal{H}_{1d}$ with eigenstates ϕ^n_i whose energies are $(m_n, -m_n, -m_n, m_n)$ for $i = 1, 2, 3, 4$ and $n = 1, \cdots, L_z$. Notice that the sequence of basis has been assigned so that $i = 1, 2$

states and $i = 3, 4$ states are separated by mirror symmetry. This procedure is general, depending only on the specific mirror or parity symmetry the system owns and not on specific parameter choices.

(iii) Project the film Hamiltonian and possible external field terms using Φ_i^n as

$$(H_{\text{equ}})_{ij}^{nn'} = \langle \phi_i^n | \, H_{\text{Film}} + V_{\text{ext}} \, | \phi_j^{n'} \rangle \, . \tag{19}$$

Basically, the projection of the original film Hamiltonian leads to

$$\tilde{H}_{\text{Film}}(\boldsymbol{k}) = \bigoplus_{n=1}^{L_z} \left[\lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + m_n(\boldsymbol{k})\tau_z\sigma_z \right], \tag{20}$$

where $2L_z$ Dirac fermions $\tilde{H}_{\text{Film}} = \oplus_{n,\chi} h_{n,\chi}(\boldsymbol{k})$ emerge as

$$h_{n,\chi}(\boldsymbol{k}) = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \chi m_n(\boldsymbol{k})\sigma_z, \tag{21}$$

with $\chi = \pm$ labelling the mirror eigenvalue.

Clearly, $\mathcal{M}_0(\boldsymbol{k})$ in $\mathcal{H}_{1d}(\boldsymbol{k})$ would be the main element dominating whether it has zero modes solution, under suitable boundary condition. In the numerical simulation, we set the model parameters to be fixed as $\lambda_{\parallel} = 0.41$eV, $\lambda_{\perp} = 0.44$eV, $t_{\parallel} = 0.566$eV, $t_{\perp} = 0.4$eV, $m_0 = 0.28$ eV, $a = b = 1$nm, $c = 0.5$nm[4].

For a semi-magnetic topological insulator attached with s-wave superconductor, the external field contains two parts:

• The Zeeman field

$$\mathcal{V}_Z(\boldsymbol{k}) = \sum_{l_z,\boldsymbol{k}} \psi_{l_z,\boldsymbol{k}}^{\dagger} V_Z(l_z) \psi_{l_z,\boldsymbol{k}}, \; V_Z(l_z) = V_z(l_z)\sigma_z\tau_0; \tag{22}$$

• The proximate superconducting pairing (simplest form)

$$\begin{aligned} \mathcal{V}_{SC}(\boldsymbol{k}) &= \sum_{l_z,\boldsymbol{k}} \Delta_k(l_z)(c_{l_z,\boldsymbol{k}\uparrow}^{1\dagger} c_{l_z,-\boldsymbol{k}\downarrow}^{1\dagger} + c_{l_z,\boldsymbol{k}\uparrow}^{2\dagger} c_{l_z,-\boldsymbol{k}\downarrow}^{2\dagger}) + \text{h.c.} \\ &= \frac{1}{2} \sum_{l_z,\boldsymbol{k}} \psi_{l_z,\boldsymbol{k}}^{\dagger} \begin{pmatrix} & \Delta_k(l_z)\tau_0 \\ -\Delta_{-k}(l_z)\tau_0 & \end{pmatrix} \psi_{l_z,-\boldsymbol{k}}^{\dagger}, \end{aligned} \tag{23}$$

which will become, by assuming s-wave pairing so that $\Delta_k = \Delta_{-k}$,

$$\mathcal{V}_{SC}(\boldsymbol{k}) = \frac{1}{2} \sum_{l_z,\boldsymbol{k}} \psi_{l_z,\boldsymbol{k}}^{\dagger} \Delta_k(l_z)(i\sigma_y)\tau_0 \psi_{l_z,-k}^{\dagger}. \tag{24}$$

From here, we already see that two choices basis can be made: absorbing the $i\sigma_y$ matrix into pairing or into Fermionic operator. Since in the latter case, the chosen pairing will contain identity spinor matrix, we choose this convention.

Now we choose Nambu basis

$$\Phi_k = \begin{pmatrix} \psi_k \\ (i\sigma_y)\psi^\dagger_{-k} \end{pmatrix}, \ \psi_k = \oplus_{l_z}\psi_{l_z,k}, \tag{25}$$

to write down the Bogoliubov-de-Gennes Hamiltonian as

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_{\text{Film}} + \mathcal{V}_Z + \mathcal{V}_{SC} = \frac{1}{2}\sum_{\boldsymbol{k}} \Phi^\dagger_{\boldsymbol{k}} H_{\text{BdG}} \Phi_{\boldsymbol{k}} \\ &= \frac{1}{2}\sum_{\boldsymbol{k}} \begin{pmatrix} \psi^\dagger_{\boldsymbol{k}} & \psi_{-k}(-i\sigma_y) \end{pmatrix} \begin{pmatrix} H_{\text{Film}}(\boldsymbol{k}) + V_Z & \Delta_k \sigma_0\tau_0 \\ \Delta^*_k\sigma_0\tau_0 & -\sigma_y[H_{\text{Film}}(-\boldsymbol{k}) + V_Z]^T\sigma_y \end{pmatrix} \begin{pmatrix} \psi_k \\ (i\sigma_y)\psi^\dagger_{-k} \end{pmatrix}, \end{aligned} \tag{26}$$

where quantities should be understood as

$$V_Z = \oplus_{l_z} V_Z(l_z), \ \Delta_k = \oplus_{l_z}\Delta_k(l_z), \tag{27}$$

and there should be an explicit chemical potential term $-\mu\sigma_0\tau_0$ added to the film Hamiltonian. Since it is easy to verify that

$$\sigma_y[H_{\text{Film}}(-\boldsymbol{k}) + V_Z]^T\sigma_y = H_{\text{Film}}(\boldsymbol{k}) - V_Z, \tag{28}$$

one concludes that

$$H_{\text{BdG}}(\boldsymbol{k}) = \begin{pmatrix} H_{\text{Film}}(\boldsymbol{k}) + V_Z & \Delta_k\sigma_0\tau_0 \\ \Delta^*_k\sigma_0\tau_0 & -H_{\text{Film}}(\boldsymbol{k}) + V_Z \end{pmatrix}, \tag{29}$$

which serves as the starting point for analysis.

Before start, let us consider once more about symmetries own by the system. Without V_Z, the system owns time-reversal symmetry $\mathcal{T} = i\sigma_y\mathcal{K}$ if and only if Δ_k is real. Particle-hole symmetry switches particle and hole, together with inverting the momentum. Generally, we write

$$\mathcal{C}\psi_k\mathcal{C}^{-1} = (\psi^\dagger_{-k}U^\dagger)^T, \ \mathcal{C}\psi^\dagger_k\mathcal{C}^{-1} = (U\psi_{-k})^T, \tag{30}$$

where U represents inner degrees of freedom mixture, while ^{T} is transpose used to make sense of the equality. Then for general Hamiltonian $\mathcal{H} = \sum_k \psi^\dagger_k h(k)\psi_k$, particle-hold symmetry

requires that

$$\begin{aligned}\mathcal{C}\mathcal{H}\mathcal{C}^{-1} &= \sum_k \mathcal{C}\psi_k^\dagger \mathcal{C}^{-1}\mathcal{C}h(k)\mathcal{C}^{-1}\mathcal{C}\psi_k\mathcal{C}^{-1} \\ &= \sum_k \psi_{-k}^T U^T h(k) U^*(\psi_{-k}^\dagger)^T = -\sum_k \psi_k^\dagger U^\dagger h^T(-k)U\psi_k + \mathrm{Tr}(H) = \mathcal{C},\end{aligned}$$

which leads to requirement due to hermicity

$$U^\dagger h^T(-k)U = -h(k),\ \mathrm{Tr}(h) = 0. \tag{31}$$

Notice that we have chosen particle-hole operator to be unitary. Under BdG representation, it is natural to choose $U = \varsigma_y\sigma_y$ as Nambu degrees of freedom which switches creation and annihilation operators

$$(\Phi_{-k}^\dagger U^\dagger)^T = \left[\begin{pmatrix}\psi_{-k}^\dagger & \psi_k(-i\sigma_y)\end{pmatrix}\begin{pmatrix} & -i\sigma_y \\ i\sigma_y & \end{pmatrix}\right]^T = \begin{pmatrix}\psi_k \\ \psi_{-k}^\dagger(-i\sigma_y)\end{pmatrix} = \begin{pmatrix}\psi_k \\ i\sigma_y\psi_{-k}^\dagger\end{pmatrix}, \tag{32}$$

and under the representation we could let $C = \varsigma_y\sigma_y\mathcal{K}$ with $\mathcal{K}$ complex-conjugate operation so that

$$Ch(k)C^{-1} = -h(-k), \tag{33}$$

and it can be verified that out H_{BdG} satisfies this by

$$\begin{aligned}U^\dagger H_{\mathrm{BdG}}^T(-k)U &= \begin{pmatrix} & -i\sigma_y \\ i\sigma_y & \end{pmatrix}\begin{pmatrix} H_{\mathrm{Film}}^T(-k) & \Delta_{-k}^* \\ \Delta_{-k} & -H_{\mathrm{Film}}^T(-k)\end{pmatrix}\begin{pmatrix} & -i\sigma_y \\ i\sigma_y & \end{pmatrix} \\ &= \begin{pmatrix} -\sigma_y H_{\mathrm{Film}}^T(-k)\sigma_y & -\sigma_y\Delta_{-k}\sigma_y \\ -\sigma_y\Delta_{-k}^*\sigma_y & \sigma_y H_{\mathrm{Film}}^T(-k)\sigma_y\end{pmatrix} = -H_{\mathrm{BdG}}(k).\end{aligned}$$

Still, our strategy is to solve the $m_0(\boldsymbol{k})$-dependent 1-D Hamiltonian inside, and now it becomes

$$H_{\mathrm{BdG}}^{1d}(\boldsymbol{k}) = \begin{pmatrix} H_{1d}(\boldsymbol{k}) & \\ & -H_{1d}(\boldsymbol{k})\end{pmatrix}, \tag{34}$$

which owns both time-reversal, particle hole above and their combination as emergent chiral symmetry. However, by the highly symmetric form, the 1-D Hamiltonian owns an additional chiral symmetry τ_y which relates its basic solutions

$$\chi(s) = i\tau_y\varphi(s),\ s = \pm, \tag{35}$$

where

$$\varphi(s,l_z,\boldsymbol{k}) = C\begin{pmatrix} -is\lambda_\perp f_+(l_z,\boldsymbol{k}) \\ t_\perp \eta(\boldsymbol{k}) f_-(l_z,\boldsymbol{k}) \end{pmatrix}, \tag{36}$$

with parity even/odd functions $f_\pm$ as

$$\begin{cases} f_+(l_z) = \dfrac{\cos(\xi_1 l_z)}{\cos(\xi_1 l/2)} - \dfrac{\cos(\xi_2 l_z)}{\cos(\xi_2 l/2)} \\ f_-(l_z) = \dfrac{\sin(\xi_1 l_z)}{\sin(\xi_1 l/2)} - \dfrac{\sin(\xi_2 l_z)}{\sin(\xi_2 l/2)} \end{cases}, \tag{37}$$

where $l = L_z + 1$ with L_z the layer number, and the dependence on wavevector is introduced by $\xi_\alpha(\boldsymbol{k}),\ \alpha = 1,2$, which again, can be self-consistently obtained by solving simultaneous equations

$$m = M_0(\boldsymbol{k}) + 2t_\perp \frac{\cos\xi_1 g(\xi_1) - \cos\xi_2 g(\xi_2)}{g(\xi_1) - g(\xi_2)}, \tag{38a}$$

$$\cos\xi_\alpha = \frac{-M_0(\boldsymbol{k})t_\perp + (-1)^{\alpha-1}\sqrt{M_0^2(\boldsymbol{k})t_\perp^2 - (t_\perp^2 - \lambda_\perp^2/4)(M_0^2(\boldsymbol{k}) + \lambda_\perp^2 - m^2)}}{2(t_\perp^2 - \lambda_\perp^2/4)}, \tag{38b}$$

with m eigenvalue of H_{1d} with defined function

$$g(\xi) = \frac{\tan(\xi l)/2}{\sin\xi}. \tag{39}$$

The factor η reads

$$\eta = \frac{-2(\cos\xi_1 - \cos\xi_2)}{\sin\xi_1\cot(\xi_1 l/2) - \sin\xi_2\cot(\xi_2 l/2)}. \tag{40}$$

The solutions of $H^{1d}_{\mathrm{BdG}}(\boldsymbol{k})$ can be constructed easily. By denoting solutions of $H_{1d}(\boldsymbol{k})$ as

$$\phi_1 = \begin{pmatrix} \varphi(+) \\ 0 \end{pmatrix},\ \phi_2 = \begin{pmatrix} 0 \\ \chi(-) \end{pmatrix},\ \phi_3 = \begin{pmatrix} \chi(+) \\ 0 \end{pmatrix},\ \phi_4 = \begin{pmatrix} 0 \\ \varphi(-) \end{pmatrix}, \tag{41}$$

whose eigenvalues are $(m,-m,-m,m)$, the solution of $H^{1d}_{\mathrm{BdG}}(\boldsymbol{k})$ can be assigned as

$$\Phi_{1\sim4} = \begin{pmatrix} \phi_{1\sim4} \\ 0 \end{pmatrix},\ \Phi_{5\sim8} = \begin{pmatrix} 0 \\ \phi_{1\sim4} \end{pmatrix}, \tag{42}$$

with eigenvalues $(m,-m,-m,m,-m,m,m,-m)$, respectively. In principle $m = m_n(\boldsymbol{k})$ with $n = 1,2,\cdots,L_z$, denoting L_z solutions, and for our purpose to focus on the lowest band, we let

$$m = m_1(\boldsymbol{k}) = \Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k}), \tag{43}$$

which reveals the surface state mass of a strong topological insulator.

The projection procedure then follows that $(H_{\rm Pro})_{ij} = \langle\Phi_i|H_{\rm BdG}|\Phi_j\rangle$, and the film and Zeeman parts are projected to be

$$\begin{aligned}&\langle\Phi|\begin{pmatrix}H_{\rm Film}(\boldsymbol{k})+V_Z & \\ & -H_{\rm Film}(\boldsymbol{k})+V_Z\end{pmatrix}|\Phi\rangle\\ &=\lambda_\parallel\varsigma_z\tau_0(\sin(k_xa)\sigma_x+\sin(k_yb)\sigma_y)-\mu\varsigma_z\tau_0\sigma_0+m(\boldsymbol{k})\varsigma_z\tau_z\sigma_z+I_S(\boldsymbol{k})\varsigma_0\tau_0\sigma_z-I_A(\boldsymbol{k})\varsigma_0\tau_y\sigma_z.\end{aligned} \tag{44}$$

To project the superconducting part out, we note that in ς space, the superconducting term is purely off-diagonal, then the only conserving projection happens between $1\sim4$ to $5\sim8$, where two quantities appear as

$$\Delta_S=|C|^2\sum_{l_z}\Delta_k(l_z)\left[\lambda_\perp^2|f_+|^2+t_\perp\eta^2|f_-|^2\right], \tag{45a}$$

$$\Delta_A=|C|^2\sum_{l_z}\Delta_k(l_z)2\lambda_\perp t_\perp\eta\,\mathrm{Re}(f_+^*f_-), \tag{45b}$$

assigned as

$$\langle\Phi_n|\Delta|\Phi_{n+4}\rangle=\langle\phi_n|\Delta_k|\phi_n\rangle=\Delta_S,\ n=1,2,3,4, \tag{46a}$$

$$\langle\Phi_n|\Delta|\Phi_{n+6}\rangle=i\Delta_A,\ n=1,2. \tag{46b}$$

The projected superconducting Hamiltonian will be

$$\begin{aligned}\langle\Phi|\Delta|\Phi\rangle&=\begin{pmatrix}&\Delta_S\tau_0\sigma_0-\Delta_A\tau_y\sigma_0\\ \text{h.c.}&\end{pmatrix}\\ &=\mathrm{Re}\,\Delta_S\varsigma_x\tau_0\sigma_0-\mathrm{Re}\,\Delta_A\varsigma_x\tau_y\sigma_0-\mathrm{Im}\,\Delta_S\varsigma_y\tau_0\sigma_0+\mathrm{Im}\,\Delta_A\varsigma_y\tau_y\sigma_0.\end{aligned} \tag{47}$$

The projected total Hamiltonian will become

$$\begin{aligned}H_{\rm Pro}&=\langle\Phi|H_{\rm BdG}|\Phi\rangle\\ &=\lambda_\parallel\varsigma_z\tau_0(\sin(k_xa)\sigma_x+\sin(k_yb)\sigma_y)-\mu\varsigma_z\tau_0\sigma_0+m(\boldsymbol{k})\varsigma_z\tau_z\sigma_z+I_S(\boldsymbol{k})\varsigma_0\tau_0\sigma_z-I_A(\boldsymbol{k})\varsigma_0\tau_y\sigma_z\\ &\quad+\mathrm{Re}\,\Delta_S\varsigma_x\tau_0\sigma_0-\mathrm{Re}\,\Delta_A\varsigma_x\tau_y\sigma_0-\mathrm{Im}\,\Delta_S\varsigma_y\tau_0\sigma_0+\mathrm{Im}\,\Delta_A\varsigma_y\tau_y\sigma_0.\end{aligned} \tag{48}$$

Written more compactly,

$$\begin{aligned}H_{\rm Pro}&=\lambda_\parallel\varsigma_z(\sin(k_xa)\sigma_x+\sin(k_yb)\sigma_y)-\mu\varsigma_z+m(\boldsymbol{k})\varsigma_z\tau_z\sigma_z+[I_S(\boldsymbol{k})\tau_0-I_A(\boldsymbol{k})\tau_y]\sigma_z\\ &\quad+\varsigma_x(\mathrm{Re}\,\Delta_S\tau_0-\mathrm{Re}\,\Delta_A\tau_y)-\varsigma_y(\mathrm{Im}\,\Delta_S\tau_0-\mathrm{Im}\,\Delta_A\tau_y).\end{aligned} \tag{49}$$

III. Basis Consideration

The above utilizes mirror-symmetric basis, whose physics is deeper but may be lack of intuitive understanding. Nevertheless, we can introduce a unitary transformation $U = U_1 U_2$, where

$$U_1 = e^{-i\tau_y\sigma_z\pi/4}, \tag{50a}$$

$$U_2 = e^{-i\tau_x\sigma_0\pi/4}, \tag{50b}$$

so that we can transform projected Hamiltonian $H_4 = \langle\phi|H_{\text{Film}} + V_Z|\phi\rangle$ without superconducting part into

$$\begin{aligned} H_{SS} &= U^\dagger H_4 U = \lambda_\parallel\tau_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - m(\boldsymbol{k})\tau_x\sigma_0 + I_S(\boldsymbol{k})\tau_0\sigma_z + I_A(\boldsymbol{k})\tau_z\sigma_z \\ &= \begin{pmatrix} \lambda_\parallel(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) + I_+(\boldsymbol{k})\sigma_z & -m(\boldsymbol{k})\sigma_0 \\ -m(\boldsymbol{k})\sigma_0 & -\lambda_\parallel(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) + I_-(\boldsymbol{k})\sigma_z \end{pmatrix}, \end{aligned} \tag{51}$$

with chemical potential ignored and $I_\pm = I_S \pm I_A$. This transformation brings the Hamiltonian into the surface-state representation labelled by τ, and clearly the low-energy Zeeman gap is directly described by $I_\pm$. Act the same transformation $U \to \varsigma_0 U$ on the projected BdG Hamiltonian by noticing that

$$U^\dagger\varsigma_{x/y}\tau_0\sigma_0 U = \varsigma_{x/y}, \tag{52a}$$

$$U^\dagger\varsigma_{x/y}\tau_y\sigma_0 U = -\varsigma_{x/y}\tau_z, \tag{52b}$$

we obtain the transformed Hamiltonian as

$$
\begin{aligned}
H_{\rm Pro}^{SS} &= U^\dagger H_{\rm Pro} U \\
&= \lambda_\parallel \varsigma_z \tau_z (\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - m(\boldsymbol{k})\varsigma_z\tau_x\sigma_0 - \mu\varsigma_z + \varsigma_0[I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z \\
&\quad + (\varsigma_x \,\mathrm{Re} - \varsigma_y \,\mathrm{Im})(\Delta_S(\boldsymbol{k})\tau_0 + \Delta_A(\boldsymbol{k})\tau_z)\sigma_0 \\
&= \begin{pmatrix}
\lambda_\parallel \tau_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) & & & \\
\quad -m(\boldsymbol{k})\tau_x\sigma_0 - \mu & & (\Delta_S(\boldsymbol{k})\tau_0 + \Delta_A(\boldsymbol{k})\tau_z)\sigma_0 & \\
\quad +[I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z & & & \\
& & -\lambda_\parallel \tau_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) & \\
(\Delta_S^*(\boldsymbol{k})\tau_0 + \Delta_A^*(\boldsymbol{k})\tau_z)\sigma_0 & & \quad +m(\boldsymbol{k})\tau_x\sigma_0 + \mu & \\
& & \quad +[I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z &
\end{pmatrix} \\
&= \begin{pmatrix}
\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z - \mu & -m & \Delta_+ & \\
-m & -\lambda_\parallel(k\times\sigma)_z + I_-\sigma_z - \mu & & \Delta_- \\
\Delta_+^* & & -\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z + \mu & m \\
& \Delta_-^* & m & \lambda_\parallel(k\times\sigma)_z + I_-\sigma_z + \mu
\end{pmatrix},
\end{aligned}
\tag{53}
$$

where in the last line, we use the continuum marks to represent the Hamiltonian for simplicity. On the other hand, we may write by putting surface index at the outside as

$$
\begin{aligned}
H_{\rm Pro}^{SS} &= \lambda_\parallel \tau_z \varsigma_z (\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - m(\boldsymbol{k})\tau_x\varsigma_z - \mu\varsigma_z + [I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z \\
&\quad + (\Delta_S(\boldsymbol{k})\tau_0 + \Delta_A(\boldsymbol{k})\tau_y)(\overleftarrow{\mathrm{Re}}\,\varsigma_x - \overleftarrow{\mathrm{Im}}\,\varsigma_y) \\
&= \begin{pmatrix}
\lambda_\parallel\varsigma_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - \mu\varsigma_z & & -m(\boldsymbol{k})\varsigma_z\sigma_0 & \\
+I_+(\boldsymbol{k})\varsigma_0\sigma_z + \Delta_+(\boldsymbol{k})(\overleftarrow{\mathrm{Re}}\,\varsigma_x - \overleftarrow{\mathrm{Im}}\,\varsigma_y)\sigma_0 & & & \\
& & -\lambda_\parallel\varsigma_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - \mu\varsigma_z & \\
-m(\boldsymbol{k})\varsigma_z\sigma_0 & & +I_-(\boldsymbol{k})\varsigma_0\sigma_z + \Delta_-(\boldsymbol{k})(\overleftarrow{\mathrm{Re}}\,\varsigma_x - \overleftarrow{\mathrm{Im}}\,\varsigma_y)\sigma_0 &
\end{pmatrix} \\
&= \begin{pmatrix}
\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z - \mu & \Delta_+ & -m & \\
\Delta_+^* & -\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z + \mu & & m \\
-m & & -\lambda_\parallel(k\times\sigma)_z + I_-\sigma_z - \mu & \Delta_- \\
& m & \Delta_-^* & \lambda_\parallel(k\times\sigma)_z + I_-\sigma_z + \mu
\end{pmatrix},
\end{aligned}
\tag{54}
$$

with physical interpretation as: left-top blocks represent Hamiltonian for the **top** surface with surface Zeeman and superconducting terms, and right-bottom for **bottom** surface;

terms off-diagonal represent top-bottom coupling, which vanishes at low-energy while mixing their high-energy bulk states as a requirement of the parity symmetry.

A. Case I: Uniform Zeeman and SC-Pairing

This case equals considering a magnetic topological insulator (MTI) thin film with long coherent length superconducting pairing immersed in.

B. Case II: Top Zeeman and Bottom SC-Pairing

For the FM-TI-SC heterostructure case, considering magnetism and superconducting pairing exist only at top and bottom surfaces separately leads to a good approximation that

$$\Delta_+ = 0,\ \Delta_- = \Delta, \tag{55a}$$

$$I_+ = V,\ I_- = 0, \tag{55b}$$

so that

$$H^{SS}_{\text{Pro}} = \begin{pmatrix} \lambda_\parallel(k\times\sigma)_z + V\sigma_z - \mu & & -m & \\ & -\lambda_\parallel(k\times\sigma)_z + V\sigma_z + \mu & & m \\ -m & & -\lambda_\parallel(k\times\sigma)_z - \mu & \Delta \\ & m & \Delta^* & \lambda_\parallel(k\times\sigma)_z + \mu \end{pmatrix}. \tag{56}$$

It should be stressed that both magnetism and superconducting pairing only modifies behavior near Fermi surface, which serves as low-energy mass term at most, and the ultimate topology will be determined by the high-energy coupling induced mass m.

IV. Sau-Lutchyn-Tewari-Das Sarma proposal

The system is a 2DEG with Rashba SOC and out-of-plane Zeeman field, in proximity to an s-wave superconductor. The Hamiltonian reads

$$\begin{aligned}\mathcal{H} &= \mathcal{H}_0 + \mathcal{H}_p \\ &= \sum_k c_k^\dagger \left[\varepsilon(k) + V_z\sigma_z + \alpha(\boldsymbol{k}\times\boldsymbol{\sigma})_z\right] c_k + \Delta c_{k,\uparrow}^\dagger c_{-k,\downarrow}^\dagger + \Delta^* c_{-k,\downarrow} c_{k,\uparrow} \\ &= \frac{1}{2}\sum_k \psi_k^\dagger \begin{pmatrix} \varepsilon(k) + V_z\sigma_z + \alpha(\boldsymbol{k}\times\boldsymbol{\sigma})_z & \Delta \\ \Delta^* & -\varepsilon(k) + V_z\sigma_z - \alpha(\boldsymbol{k}\times\boldsymbol{\sigma})_z \end{pmatrix} \psi_k,\end{aligned} \tag{57}$$

where $\varepsilon(k) = \frac{k^2}{2m} - \mu$, $\psi_k = (c_k, i\sigma_y c_{-k}^\dagger)^T$. The system is topologically nontrivial by considering the Γ point spectrum, with phase boundary determined by $\det(H(0)) = 0$, i.e.,

$$\begin{vmatrix} -\mu + V_z\sigma_z & \Delta \\ \Delta^* & \mu + V_z\sigma_z \end{vmatrix} = \left(V_z^2 - \mu^2 - |\Delta|^2\right)^2 = 0. \tag{58}$$

Further considering special point in each region reveals that the non-trivial condition is

$$V_z^2 > \mu^2 + |\Delta|^2. \tag{59}$$

Introduce the unitary transformation that diagonalizes the normal part of the Hamiltonian,

$$c_k = U_k a_k \Leftrightarrow \begin{pmatrix} c_{k,\uparrow} \\ c_{k,\downarrow} \end{pmatrix} = \begin{pmatrix} \cos(\theta_k/2) & -e^{-i\varphi_k}\sin(\theta_k/2) \\ e^{i\varphi_k}\sin(\theta_k/2) & \cos(\theta_k/2) \end{pmatrix} \begin{pmatrix} a_{k,+} \\ a_{k,-} \end{pmatrix}, \tag{60}$$

we have that $U_k^\dagger H_0(k) U_k = \varepsilon(k) + \Lambda_k\sigma_z$, where $\Lambda_k = \sqrt{V_z^2 + \alpha^2 k^2}$, or equivalently,

$$\mathcal{H}_0 = \sum_{k,\nu=\pm} a_{k,\nu}^\dagger \left[\varepsilon(k) + \nu\Lambda_k\right] a_{k,\nu}, \tag{61}$$

and notice that $\cos(\theta_k) = V_z/\Lambda_k$, $\varphi_k = \arg(\boldsymbol{k}) + \pi/2$.

To see how this transformation modifies the pairing term, we can directly have

$$
\begin{aligned}
\mathcal{H}_p &= \sum_k \Delta c^\dagger_{k,\uparrow} c^\dagger_{-k,\downarrow} + \text{h.c.} \\
&= \sum_k \Delta \left(\cos\frac{\theta_k}{2} a^\dagger_{k,+} - e^{i\varphi_k} \sin\frac{\theta_k}{2} a^\dagger_{k,-} \right) \left(-e^{-i\varphi_k} \sin\frac{\theta_k}{2} a^\dagger_{-k,+} + \cos\frac{\theta_k}{2} a^\dagger_{-k,-} \right) + \text{h.c.} \\
&= \sum_k \Delta \left(-\cos\frac{\theta_k}{2} \sin\frac{\theta_k}{2} e^{-i\varphi_k} a^\dagger_{k,+} a^\dagger_{-k,+} + \cos^2\frac{\theta_k}{2} a^\dagger_{k,+} a^\dagger_{-k,-} + \sin^2\frac{\theta_k}{2} a^\dagger_{k,-} a^\dagger_{-k,+} \right. \\
&\left. - \sin\frac{\theta_k}{2} \cos\frac{\theta_k}{2} e^{i\varphi_k} a^\dagger_{k,-} a^\dagger_{-k,-} \right) + \text{h.c.} \\
&= \sum_k \left[\Delta \cos\theta_k a^\dagger_{k,+} a^\dagger_{-k,-} - \frac{1}{2} \sum_{\nu=\pm} \Delta \sin\theta_k e^{-i\nu\varphi_k} a^\dagger_{k,\nu} a^\dagger_{-k,\nu} \right] + \text{h.c.} \\
&= \frac{1}{2} \sum_k a^\dagger_k \begin{pmatrix} & & -\Delta \sin\theta_k e^{-i\varphi_k} & \Delta \cos\theta_k \\ & & -\Delta \cos\theta_k & -\Delta \sin\theta_k e^{i\varphi_k} \\ -\Delta^* \sin\theta_k e^{i\varphi_k} & -\Delta^* \cos\theta_k & & \\ \Delta^* \cos\theta_k & -\Delta^* \sin\theta_k e^{-i\varphi_k} & & \end{pmatrix} a_k,
\end{aligned}
\tag{62}
$$

where one notices that $\theta_{-k} = \theta_k$, while $e^{i\varphi_{-k}} = -e^{i\varphi_k}$, and $a^\dagger_k = \begin{pmatrix} a^\dagger_{k,+} & a^\dagger_{k,-} & a_{-k,+} & a_{-k,-} \end{pmatrix}$. It is also direct to perform unitary transformation in the BdG Hamiltonian.

The new pairing term shares an s-wave pairing part by the $+-{-}$ pairing, and a p-wave pairing part by the $+-{+}$ or $-{-}-$ pairing. It is interesting to notice that the p-wave pairing strength is proportional to $\sin\theta_k$, which becomes larger as the SOC strength increases. Now let us redefine the basic Hamiltonian as the transformed normal part plus the s-wave pairing part,

$$
\begin{aligned}
\tilde{\mathcal{H}}_0 &= \sum_k \left(\sum_\nu a^\dagger_{k,\nu} (\varepsilon(k) + \nu\Lambda_k) a_{k,\nu} \right) + \Delta \cos\theta_k a^\dagger_{k,+} a^\dagger_{-k,-} + \text{h.c.} \\
&= \frac{1}{2} \sum_k a^\dagger_k \begin{pmatrix} \varepsilon(k) + \Lambda_k & & & \Delta \cos\theta_k \\ & \varepsilon(k) - \Lambda_k & -\Delta \cos\theta_k & \\ & -\Delta^* \cos\theta_k & -\varepsilon(k) - \Lambda_k & \\ \Delta^* \cos\theta_k & & & -\varepsilon(k) + \Lambda_k \end{pmatrix} a_k \\
&= \frac{1}{2} \sum_{k,\nu=\pm} \begin{pmatrix} a^\dagger_{k,\nu} & a_{-k,-\nu} \end{pmatrix} \begin{pmatrix} \varepsilon(k) + \nu\Lambda_k & \nu\Delta \cos\theta_k \\ \nu\Delta^* \cos\theta_k & -\varepsilon(k) + \nu\Lambda_k \end{pmatrix} \begin{pmatrix} a_{k,\nu} \\ a^\dagger_{-k,-\nu} \end{pmatrix} \\
&= \frac{1}{2} \sum_{k,\nu} \alpha^\dagger_{k,\nu} \left(\nu\Lambda_k + \nu\Delta \cos\theta_k \rho_x + \varepsilon(k) \rho_z \right) \alpha_{k,\nu},
\end{aligned}
\tag{63}
$$

where in the last we assume Δ real. Thus, a Bogoliubov transformation can be performed

to diagonalize the Hamiltonian

$$\alpha_{k,\nu} = B_k \beta_{k,\nu} \Leftrightarrow \begin{pmatrix} a_{k,\nu} \\ a^\dagger_{-k,-\nu} \end{pmatrix} = \begin{pmatrix} \cos(\gamma_k/2) & -\sin(\gamma_k/2) \\ \sin(\gamma_k/2) & \cos(\gamma_k/2) \end{pmatrix} \begin{pmatrix} b_{k,\nu} \\ b^\dagger_{-k,-\nu} \end{pmatrix}, \tag{64}$$

such that

$$\tilde{\mathcal{H}}_0 = \frac{1}{2} \sum_{k,\nu} \beta^\dagger_{k,\nu} \left(\nu \Lambda_k + \sqrt{\varepsilon(k)^2 + \Delta^2 \cos^2\theta_k} \rho_z \right) \beta_{k,\nu}. \tag{65}$$

At the same time, the p-wave pairing term is transformed as

$$\begin{aligned}
\tilde{\mathcal{H}}_p &= -\frac{1}{2} \sum_{k,\nu} \Delta \sin\theta_k e^{-i\nu\varphi_k} a^\dagger_{k,\nu} a^\dagger_{-k,\nu} + \text{h.c.} \\
&= -\frac{1}{2} \sum_{k,\nu} \Delta \sin\theta_k e^{-i\nu\varphi_k} \left(\cos(\gamma_k/2) b^\dagger_{k,\nu} - \sin(\gamma_k/2) b_{-k,-\nu} \right) \\
&\quad \cdot \left(\sin(\gamma_k/2) b_{k,-\nu} + \cos(\gamma_k/2) b^\dagger_{-k,\nu} \right) + \text{h.c.} \\
&= -\frac{1}{2} \sum_{k,\nu} \Delta \sin\theta_k e^{-i\nu\varphi_k} \left(\cos^2(\gamma_k/2) b^\dagger_{k,\nu} b^\dagger_{-k,\nu} - \sin^2(\gamma_k/2) b_{-k,-\nu} b_{k,-\nu} \right) \\
&\quad - \frac{1}{4} \sum_{k,\nu} \Delta \sin\theta_k e^{-i\nu\varphi_k} \left(\sin(\gamma_k) b^\dagger_{k,\nu} b_{k,-\nu} - \sin(\gamma_k) b_{-k,-\nu} b^\dagger_{-k,\nu} \right) + \text{h.c.} \\
&= -\frac{1}{2} \sum_{k,\nu} \Delta \sin\theta_k e^{-i\nu\varphi_k} b^\dagger_{k,\nu} b^\dagger_{-k,\nu} + \text{h.c.},
\end{aligned} \tag{66}$$

where $\cos\gamma_k = \varepsilon(k)/\sqrt{\varepsilon(k)^2 + \Delta^2\cos^2\theta_k}$, $\sin\gamma_k = \Delta\cos\theta_k/\sqrt{\varepsilon(k)^2 + \Delta^2\cos^2\theta_k}$. Notice how we introduce the transformation, as they belong to two sectors. Notice how ν and h.c. play their role in the last equality.

The final Hamiltonian reads, by correctly considering the terms in order

$$\begin{aligned}
\mathcal{H} &= \sum_k \beta^\dagger_k \begin{pmatrix} \Lambda_k + \tilde{\varepsilon}_k & & -\Delta \sin\theta_k e^{-i\varphi_k} & \\ & -\Lambda_k + \tilde{\varepsilon}_k & & -\Delta \sin\theta_k e^{i\varphi_k} \\ -\Delta^* \sin\theta_k e^{i\varphi_k} & & -\Lambda_k - \tilde{\varepsilon}_k & \\ & -\Delta^* \sin\theta_k e^{-i\varphi_k} & & \Lambda_k - \tilde{\varepsilon}_k \end{pmatrix} \beta_k \\
&= \sum_{k,\nu} \gamma^\dagger_{k,\nu} \left(\tilde{\Delta} (\boldsymbol{k} \times \boldsymbol{\tau}^\nu)_z + (\tilde{\varepsilon}_k + \nu\Lambda_k) \tau_z \right) \gamma_{k,\nu},
\end{aligned} \tag{67}$$

where $\tilde{\varepsilon}_k = \sqrt{\varepsilon(k)^2 + \Delta^2\cos^2\theta_k}$, $\boldsymbol{\tau}^\nu = (\tau_x, \nu\tau_y)$, and $\tilde{\Delta} = \alpha\Delta/\Lambda_k$ with $\gamma_{k,\nu} = \begin{pmatrix} b_{k,\nu} \\ b^\dagger_{-k,\nu} \end{pmatrix}$.

Marvelously, the Hamiltonian can be written in a form similar to the Green-Read p-wave superconductor Hamiltonian, with two copies. In each copy, the mass term shares a fixed

sign at its high energy limit (this may be problematic) caused by the $\varepsilon(k)$ term, and to achieve non-triviality, the low energy mass term should be negative, which is possible only for $\nu = -1$:

$$-\Lambda_0 + \tilde{\varepsilon}_0 < 0 \Longrightarrow \Delta^2 + \mu^2 < V_z^2. \tag{68}$$

This gives the correct topological boundary condition for the system, and the superconducting Chern number is

$$\begin{cases} N_+ = -\frac{1}{2}(\mathrm{sgn}(|V_z| + \sqrt{\Delta^2 + \mu^2}) - 1) = 0 \\ N_- = -\frac{1}{2}(\mathrm{sgn}(-|V_z| + \sqrt{\Delta^2 + \mu^2}) - 1) = \begin{cases} 1, & \Delta^2 + \mu^2 < V_z^2 \\ 0, & \Delta^2 + \mu^2 > V_z^2 \end{cases} \end{cases}. \tag{69}$$

Notice that we can directly perform the transformation in the BdG Hamiltonian. By writing

$$H_{\mathrm{BdG}}(\boldsymbol{k}) = \begin{pmatrix} \varepsilon(k) + V_z\sigma_z + \alpha(\boldsymbol{k}\times\boldsymbol{\sigma})_z & \Delta \\ \Delta^* & -\varepsilon(k) + V_z\sigma_z - \alpha(\boldsymbol{k}\times\boldsymbol{\sigma})_z \end{pmatrix}, \tag{70}$$

we can define the first unitary transformation as

$$U_1 = \begin{pmatrix} U_+ & \\ & U_- \end{pmatrix},\ U_s = \begin{pmatrix} \cos(\theta_k/2) & -se^{-i\varphi_k}\sin(\theta_k/2) \\ se^{i\varphi_k}\sin(\theta_k/2) & \cos(\theta_k/2) \end{pmatrix}, \tag{71}$$

which leads to

$$\begin{aligned} U_1^\dagger H_{\mathrm{BdG}} U_1 &= \begin{pmatrix} \varepsilon(k)+\Lambda_k & & \Delta\cos\theta_k & \Delta e^{-i\varphi_k}\sin\theta_k \\ & \varepsilon(k)-\Lambda_k & \Delta e^{i\varphi_k}\sin\theta_k & \Delta\cos\theta_k \\ \Delta\cos\theta_k & \Delta e^{-i\varphi_k}\sin\theta_k & -\varepsilon(k)+\Lambda_k & \\ \Delta e^{i\varphi_k}\sin\theta_k & \Delta\cos\theta_k & & -\varepsilon(k)-\Lambda_k \end{pmatrix} \\ &\sim \begin{pmatrix} \varepsilon(k)+\Lambda_k & \Delta\cos\theta_k & & \Delta e^{-i\varphi_k}\sin\theta_k \\ \Delta\cos\theta_k & -\varepsilon(k)+\Lambda_k & \Delta e^{-i\varphi_k}\sin\theta_k & \\ & \Delta e^{i\varphi_k}\sin\theta_k & \varepsilon(k)-\Lambda_k & \Delta\cos\theta_k \\ \Delta e^{i\varphi_k}\sin\theta_k & & \Delta\cos\theta_k & -\varepsilon(k)-\Lambda_k \end{pmatrix} \\ &= \oplus_s \left(s\Lambda_k + \Delta\cos\theta_k\rho_x + \varepsilon(k)\rho_z\right) + H_p \end{aligned} \tag{72}$$

as easy to see. The next step is to perform the second unitary transformation, which

diagonalizes the first part of the Hamiltonian, and we have

$$
\begin{aligned}
U_2^\dagger(U_1^\dagger H_{\mathrm{BdG}}U_1)U_2 &= \begin{pmatrix} \tilde{\varepsilon}_k+\Lambda_k & & & \Delta\sin\theta_k e^{-i\varphi_k} \\ & -\tilde{\varepsilon}_k+\Lambda_k & \Delta\sin\theta_k e^{-i\varphi_k} & \\ & \Delta\sin\theta_k e^{i\varphi_k} & \tilde{\varepsilon}_k-\Lambda_k & \\ \Delta\sin\theta_k e^{i\varphi_k} & & & -\tilde{\varepsilon}_k-\Lambda_k \end{pmatrix} \\
&\sim \begin{pmatrix} \tilde{\varepsilon}_k+\Lambda_k & & & \Delta\sin\theta_k e^{-i\varphi_k} \\ & \tilde{\varepsilon}_k-\Lambda_k & \Delta\sin\theta_k e^{i\varphi_k} & \\ & \Delta\sin\theta_k e^{-i\varphi_k} & -\tilde{\varepsilon}_k+\Lambda_k & \\ \Delta\sin\theta_k e^{i\varphi_k} & & & -\tilde{\varepsilon}_k-\Lambda_k \end{pmatrix} \\
&\sim \oplus_\nu \left((\nu\tilde{\varepsilon}_k+\Lambda_k)\rho_z + \Delta\sin\theta_k(\boldsymbol{k}\times\boldsymbol{\tau}^\nu)_z\right).
\end{aligned}
\tag{73}
$$

As we have claimed, the high energy mass sign of the transformed Hamiltonian is fixed, and this is definitely a problem. This indeed originates from the first unitary transformation, which is the solution to the normal part of the Hamiltonian. The point is, to make the transformation valid in whole k-space, we have to make sure that the gauge consistency of the transformation, consisted of the wavefunctions, is satisfied in the whole k-space. For our 2-band case, we basically have two choices of the transformation:

$$
U_k^1 = \begin{pmatrix} \cos(\theta_k/2) & -e^{-i\varphi_k}\sin(\theta_k/2) \\ e^{i\varphi_k}\sin(\theta_k/2) & \cos(\theta_k/2) \end{pmatrix},\ U_k^2 = \begin{pmatrix} e^{i\varphi_k}\cos(\theta_k/2) & -\sin(\theta_k/2) \\ \sin(\theta_k/2) & e^{-i\varphi_k}\cos(\theta_k/2) \end{pmatrix}, \tag{74}
$$

where difference lies in whether the phase factor is accompanied by the cos term or the sin term. In the above calculation, we blindly choose U_k^1 to perform the transformation, and this is valid only for $V_z > 0$ case. To see why, we consider that

$$
\cos\theta_k = \frac{V_z}{\sqrt{V_z^2+\alpha^2k^2}},\ \sin\theta_k = \frac{\alpha k}{\sqrt{V_z^2+\alpha^2k^2}},\ e^{i\varphi_k} = \frac{-k_y+ik_x}{k},
$$

and we see that $\sin\theta_k$ is always positive, making $\theta_k \in [0,\pi)$ and $\theta_k/2 \in [0,\pi/2)$, consistent as the polar angle. Then we obtain

$$
\begin{aligned}
\cos\frac{\theta_k}{2} &= \sqrt{\frac{1+\cos\theta_k}{2}} = \frac{1}{\sqrt{2}}\sqrt{1+\frac{V_z}{\Lambda_k}}, \\
\sin\frac{\theta_k}{2} &= \sqrt{\frac{1-\cos\theta_k}{2}} = \frac{1}{\sqrt{2}}\sqrt{1-\frac{V_z}{\Lambda_k}},
\end{aligned}
$$

with $\Lambda_k = \sqrt{V_z^2 + \alpha^2 k^2}$. The phase $e^{i\varphi_k}$ raises the question about the $k = 0$ point, since

$$\lim_{k_x \to 0} \lim_{k_y \to 0} e^{i\varphi_k} = \lim_{k_x \to 0} \frac{ik_x}{|k_x|} = i\mathrm{sgn}(k_x),$$
$$\lim_{k_y \to 0} \lim_{k_x \to 0} e^{i\varphi_k} = \lim_{k_y \to 0} \frac{-k_y}{|k_y|} = -\mathrm{sgn}(k_y),$$

which indicates that the phase is not continuous at $k = 0$ point. In the SLTD proposal, the $k = 0$ point is considered to be occupied for chemical potential lying inside the Zeeman gap, and its gauge consistency is essential, and we have to remove this singularity. This can be achieved by noticing that

$$\cos\frac{\theta_0}{2} = \frac{1 + \mathrm{sgn}(V_z)}{\sqrt{2}},\ \sin\frac{\theta_0}{2} = \frac{1 - \mathrm{sgn}(V_z)}{\sqrt{2}},$$

then as long as $V_z \neq 0$, we can always assign $e^{i\varphi_k}$ to $\sin\frac{\theta_k}{2}$ when $V_z > 0$, and to $\cos\frac{\theta_k}{2}$ when $V_z < 0$, such that the singularity is removed.

Now we need to consider the $V_z < 0$ case, where we have to use U_k^2 to perform the transformation. The key point is, the emergent p-wave pairing is accompanied with the $\sin\theta_k = 2\cos\frac{\theta_k}{2}\sin\frac{\theta_k}{2}$ factor, which contains both $\cos\frac{\theta_k}{2}$ and $\sin\frac{\theta_k}{2}$ terms. Thus, no matter to which term we assign the phase factor it always appear in the p-wave pairing — but for two gauges, one is $e^{i\varphi_k}$ and the other is $e^{-i\varphi_k}$, leading to the opposite 'chirality' of the p-wave pairing. We thus obtain the final Hamiltonian for $V_z < 0$ case as

$$\mathcal{H} = \sum_{k,\nu} \gamma_{k,\nu}^\dagger \left(\tilde{\Delta}(\boldsymbol{k} \times \boldsymbol{\tau}^{-\nu})_z + (\tilde{\varepsilon}_k + \nu\Lambda_k)\tau_z \right) \gamma_{k,\nu}, \tag{75}$$

where $\tilde{\varepsilon}_k = \sqrt{\varepsilon(k)^2 + \Delta^2\cos^2\theta_k}$, $\boldsymbol{\tau}^{-\nu} = (\tau_x, -\nu\tau_y)$, and $\tilde{\Delta} = \alpha\Delta/\Lambda_k$ with $\gamma_{k,\nu} = \begin{pmatrix} b_{k,\nu} \\ b_{-k,\nu}^\dagger \end{pmatrix}$. Notice that it is still the $\nu = -1$ case that gives non-trivial topology, but due to the opposite chirality of the p-wave pairing, the Chern number is now positive as

$$\begin{cases} N_+ = \frac{1}{2}(\mathrm{sgn}(|V_z| + \sqrt{\Delta^2 + \mu^2}) - 1) = 0 \\ N_- = \frac{1}{2}(\mathrm{sgn}(-|V_z| + \sqrt{\Delta^2 + \mu^2}) - 1) = \begin{cases} -1, & \Delta^2 + \mu^2 < V_z^2 \\ 0, & \Delta^2 + \mu^2 > V_z^2 \end{cases} \end{cases}. \tag{76}$$

Finally, we can summarize the results as follows:

$$\mathcal{H}^{V_z} = \sum_{k,\nu} \gamma_{k,\nu}^\dagger \left(\tilde{\Delta}(\boldsymbol{k} \times \boldsymbol{\tau}^{\mathrm{sgn}(V_z)\nu})_z + (\tilde{\varepsilon}_k + \nu\Lambda_k)\tau_z \right) \gamma_{k,\nu}, \tag{77}$$

$$\begin{cases} N_+^{V_z} = -\mathrm{sgn}(V_z)\frac{1}{2}(\mathrm{sgn}(|V_z| + \sqrt{\Delta^2+\mu^2}) - 1) = 0 \\ N_-^{V_z} = -\mathrm{sgn}(V_z)\frac{1}{2}(\mathrm{sgn}(-|V_z| + \sqrt{\Delta^2+\mu^2}) - 1) = \begin{cases} \mathrm{sgn}(V_z), & \Delta^2+\mu^2 < V_z^2 \\ 0, & \Delta^2+\mu^2 > V_z^2 \end{cases} \end{cases} . \quad (78)$$

V. Qi-Hughes-Zhang Model

In the early proposal[9], people directly employee a two band model

$$h_0 = \lambda(\boldsymbol{k}\cdot\boldsymbol{\sigma}) + m(\boldsymbol{k})\sigma_z, \quad (79)$$

where the mass term $m(\boldsymbol{k}) = m_0 - tk^2$, to illustrate the formation of p-wave superconductivity. In proximity with an s-wave superconductor, the BdG Hamiltonian is given by

$$H_{\mathrm{BdG}} = \rho_z(\lambda\boldsymbol{k}\cdot\boldsymbol{\sigma} - \mu) + m(\boldsymbol{k})\sigma_z + \Delta\rho_x, \quad (80)$$

and a p-wave chiral topological superconductor is formed when the following condition is satisfied

$$\sqrt{\Delta^2+\mu^2} > m_0^2. \quad (81)$$

Intuitively, when this condition is not satisfied, the system is either adiabatically connected to an trivial insulator with superconducting Chern number zero, or a Chern insulator with one Fermionic edge state, or equivalently, two Majorana edge modes with superconducting Chern number two. It is only when the condition is satisfied that the system is a chiral p-wave topological superconductor with superconducting Chern number one, where a single gapless chiral Majorana edge mode is formed at the edge of the system.

Be cautious that in the early proposal, people highly bind the concept of quantum (anomalous) Hall effect[10, 11] (QHE or QAHE) with the topological superconductivity, which, however, as we shall show later, is not the case. It is indeed rather clear from the criterion itself that as the pairing gap Δ is small, which is the usual case for a conventional superconductor, the system is a topological superconductor when $|\mu| > |m_0|$, i.e., when the normal state system is a metal, with non-vanishing anomalous Hall conductivity

$$\sigma_H = \frac{e^2}{2h}\left[\mathrm{sgn}(t) - \frac{m(k_F)}{\sqrt{\lambda^2 k_F^2 + m^2(k_F)}}\right], \quad (82)$$

with k_F the Fermi momentum determined by $\mu^2 = \lambda^2 k_F^2 + m^2(k_F)$. Notice that as $|\mu| > |m_0|$, we always solve a non-vanishing

$$\begin{aligned} k_F^2 &= \tilde{\lambda}^2 \left(\sqrt{1 - 2\tilde{m}_0 + \tilde{\mu}^2} - 1 + \tilde{m}_0 \right) \\ &= \tilde{\lambda}^2 |1 - \tilde{m}_0| \left(\sqrt{1 + \frac{\tilde{\mu}^2 - \tilde{m}_0^2}{(1 - \tilde{m}_0)^2}} - \mathrm{sgn}(1 - \tilde{m}_0) \right), \end{aligned} \tag{83}$$

where $\tilde{\lambda} = \lambda/\sqrt{2}t$, $\tilde{m}_0 = 2m_0 t/\lambda^2$, $\tilde{\mu} = 2\mu t/\lambda^2$, indicating the existence of a Fermi surface.

To see the p-wave nature, we similarly introduce

$$U_s^1 = \begin{pmatrix} \cos(\theta_k/2) & -se^{-i\varphi_k}\sin(\theta_k/2) \\ se^{i\varphi_k}\sin(\theta_k/2) & \cos(\theta_k/2) \end{pmatrix}, \; U_s^2 = \begin{pmatrix} se^{i\varphi_k}\cos(\theta_k/2) & -\sin(\theta_k/2) \\ \sin(\theta_k/2) & se^{-i\varphi_k}\cos(\theta_k/2) \end{pmatrix}, \tag{84}$$

where $\cos\theta_k = m(k)/\Lambda_k$, with $\Lambda_k = \sqrt{\lambda^2 k^2 + m^2(k)}$, $e^{i\varphi_k} = (k_x + ik_y)/k$. As

$$\cos\frac{\theta_k}{2} = \frac{1}{\sqrt{2}}\sqrt{1 + \frac{m(k)}{\Lambda_k}}, \; \sin\frac{\theta_k}{2} = \frac{1}{\sqrt{2}}\sqrt{1 - \frac{m(k)}{\Lambda_k}},$$

we again face the gauge choosing problem. At this time, different from the SLTD model where the sign of V_z matters as the low-energy $k = 0$ area is occupied, here it is the $k \to \infty$ limit that matters, since the Fermi sea goes to infinity. Since we have

$$\lim_{k\to\infty} \frac{m(k)}{\Lambda_k} = -\mathrm{sgn}(t), \tag{85}$$

we need to choose U_s^2 when $t > 0$ and U_s^1 when $t < 0$.

People may worry about the case when $m(k)$ leads to a non-trivial band structure, i.e., $m_0 t > 0$ such that we can not extend either U_s^1 or U_s^2 to the whole k-space. Our answer will be: yes, we can not extend the transformation to the whole k-space mathematically, but we can do that physically. This is because, superconductor is about Fermi surface, and topology is about ground state wavefunction. The ground state wavefunction of the proximate system originates from the normal state wavefunction, which is a Fermi sea, and cares only about the occupied states. We thus extend the transformation to the whole k-space formally.

Since the Hamiltonian has a similar form, we have

$$U_1 = \begin{pmatrix} U_+ & \\ & U_- \end{pmatrix}, \tag{86}$$

which leads to

$$
\begin{aligned}
U_1^\dagger H_{\mathrm{BdG}} U_1 &= \begin{pmatrix}
-\mu+\Lambda_k & & \Delta\cos\theta_k & \Delta e^{i\mathrm{sgn}(t)\varphi_k}\sin\theta_k \\
& -\mu-\Lambda_k & \Delta e^{-i\mathrm{sgn}(t)\varphi_k}\sin\theta_k & \Delta\cos\theta_k \\
\Delta\cos\theta_k & \Delta e^{i\mathrm{sgn}(t)\varphi_k}\sin\theta_k & \mu+\Lambda_k & \\
\Delta e^{-i\mathrm{sgn}(t)\varphi_k}\sin\theta_k & \Delta\cos\theta_k & & \mu-\Lambda_k
\end{pmatrix} \\
&\sim \begin{pmatrix}
-\mu+\Lambda_k & \Delta\cos\theta_k & & \Delta e^{i\mathrm{sgn}(t)\varphi_k}\sin\theta_k \\
\Delta\cos\theta_k & \mu+\Lambda_k & \Delta e^{i\mathrm{sgn}(t)\varphi_k}\sin\theta_k & \\
& \Delta e^{-i\mathrm{sgn}(t)\varphi_k}\sin\theta_k & -\mu-\Lambda_k & \Delta\cos\theta_k \\
\Delta e^{-i\mathrm{sgn}(t)\varphi_k}\sin\theta_k & & \Delta\cos\theta_k & \mu-\Lambda_k
\end{pmatrix} \\
&= \oplus_s \left(s\Lambda_k + \Delta\cos\theta_k\rho_x - \mu\rho_z\right) + H_p.
\end{aligned}
\tag{87}
$$

A second unitary transformation is needed to diagonalize the first part of the Hamiltonian, and we have

$$
\begin{aligned}
U_2^\dagger(U_1^\dagger H_{\mathrm{BdG}} U_1)U_2 &= \begin{pmatrix}
\tilde{\varepsilon}_k+\Lambda_k & & & \Delta\sin\theta_k e^{i\mathrm{sgn}(t)\varphi_k} \\
& -\tilde{\varepsilon}_k+\Lambda_k & \Delta\sin\theta_k e^{i\mathrm{sgn}(t)\varphi_k} & \\
& \Delta\sin\theta_k e^{-i\mathrm{sgn}(t)\varphi_k} & \tilde{\varepsilon}_k-\Lambda_k & \\
\Delta\sin\theta_k e^{-i\mathrm{sgn}(t)\varphi_k} & & & -\tilde{\varepsilon}_k-\Lambda_k
\end{pmatrix} \\
&\sim \begin{pmatrix}
\tilde{\varepsilon}_k+\Lambda_k & & & \Delta\sin\theta_k e^{i\mathrm{sgn}(t)\varphi_k} \\
& \tilde{\varepsilon}_k-\Lambda_k & \Delta\sin\theta_k e^{-i\mathrm{sgn}(t)\varphi_k} & \\
& \Delta\sin\theta_k e^{i\mathrm{sgn}(t)\varphi_k} & -\tilde{\varepsilon}_k+\Lambda_k & \\
\Delta\sin\theta_k e^{-i\mathrm{sgn}(t)\varphi_k} & & & -\tilde{\varepsilon}_k-\Lambda_k
\end{pmatrix} \\
&\sim \oplus_\nu \left((\nu\tilde{\varepsilon}_k+\Lambda_k)\rho_z + \Delta\sin\theta_k(\boldsymbol{k}\times\boldsymbol{\tau}^{-\mathrm{sgn}(t)\nu})_z\right),
\end{aligned}
\tag{88}
$$

where $\tilde{\varepsilon}_k = \sqrt{\mu^2+\Delta^2\cos^2\theta_k}$. The non-trivial phase occurs for $\nu=-1$ case, when

$$
(-\tilde{\varepsilon}_0+\Lambda_0)(-\tilde{\varepsilon}_\infty+\Lambda_\infty) < 0 \Rightarrow \mu^2+\Delta^2 > m_0^2. \tag{89}
$$

The superconducting Chern number is given by

$$
N = -\frac{\mathrm{sgn}(t)}{2}(\mathrm{sgn}(|m_0|-\sqrt{\Delta^2+\mu^2})-1) = \begin{cases} -\mathrm{sgn}(t), & \Delta^2+\mu^2 > m_0^2 \\ 0, & \Delta^2+\mu^2 < m_0^2 \end{cases}. \tag{90}
$$

VI. TI Film Model

The Hamiltonian reads

$$\begin{aligned} H_{\rm BdG} = &\lambda\rho_z\tau_0(\boldsymbol{k}\cdot\boldsymbol{\sigma}) - \mu\rho_z\tau_0\sigma_0 + m(\boldsymbol{k})\rho_z\tau_z\sigma_z + \rho_0(I_S\tau_0 - I_A\tau_y)\sigma_z \\ &+ \rho_x(\Delta_S\tau_0 - \Delta_A\tau_y)\sigma_0. \end{aligned} \tag{91}$$

The system owns a particle-hole symmetry $\mathcal{C} = \rho_y\tau_x\sigma_y\mathcal{K}$. To find its effective Hamiltonian, we need at least three steps of unitary transformations.

The first step performs the effective-mass transformation in τ space, that

$$U_M^{s,\dagger}(sm\tau_z + I_S\tau_0 - I_A\tau_y)U_M^s = \begin{pmatrix} \tilde{m}_+ & \\ & \tilde{m}_- \end{pmatrix}, \tag{92}$$

where

$$U_M^s = \frac{1}{\sqrt{2}}\begin{pmatrix} i\mathrm{sgn}(I_A)\sqrt{1+\frac{sm}{M}} & \sqrt{1-\frac{sm}{M}} \\ \sqrt{1-\frac{sm}{M}} & i\mathrm{sgn}(I_A)\sqrt{1+\frac{sm}{M}} \end{pmatrix} = \begin{pmatrix} is_A\cos\frac{\theta^s}{2} & \sin\frac{\theta^s}{2} \\ \sin\frac{\theta^s}{2} & is_A\cos\frac{\theta^s}{2} \end{pmatrix}, \tag{93}$$

with $s = \pm$ labelling ρ_z and $M(\boldsymbol{k}) = \sqrt{m^2(\boldsymbol{k}) + I_A^2(\boldsymbol{k})}$, $\cos\theta^s = s\frac{m}{M}$, $\sin\theta^s = |I_A|/M$. The effective masses $\tilde{m}_\chi = I_S + \chi M$ with $\chi = \pm$ labelling τ_z.

[1] L. Fu, C. L. Kane, and E. J. Mele, Physical review letters **98**, 106803 (2007).

[2] Y. Xia, D. Qian, D. Hsieh, L. Wray, A. Pal, H. Lin, A. Bansil, D. Grauer, Y. S. Hor, R. J. Cava, *et al.*, Nature physics **5**, 398 (2009).

[3] D. Hsieh, Y. Xia, D. Qian, L. Wray, F. Meier, J. Dil, J. Osterwalder, L. Patthey, A. Fedorov, H. Lin, *et al.*, Physical review letters **103**, 146401 (2009).

[4] H. Zhang, C.-X. Liu, X.-L. Qi, X. Dai, Z. Fang, and S.-C. Zhang, Nature physics **5**, 438 (2009).

[5] Y. Chen, J. G. Analytis, J.-H. Chu, Z. Liu, S.-K. Mo, X.-L. Qi, H. Zhang, D. Lu, X. Dai, Z. Fang, *et al.*, science **325**, 178 (2009).

[6] S.-Q. Shen, *Topological insulators*, Vol. 174 (Springer, 2017).

[7] K.-Z. Bai, B. Fu, Z. Zhang, and S.-Q. Shen, arXiv preprint arXiv:2308.05963 (2023).

[8] S.-Q. Shen, https://doi.org/10.21203/rs.3.rs-3384751/v1 (2023).

[9] X.-L. Qi, T. L. Hughes, and S.-C. Zhang, Phys. Rev. B **82**, 184516 (2010).

[10] K. v. Klitzing, G. Dorda, and M. Pepper, Phys. Rev. Lett. **45**, 494 (1980).

[11] F. D. M. Haldane, Physical review letters **61**, 2015 (1988).